High Performance Programming

”Parallel Program Paradigm and Design Process”

Ramoni Adeogun

AALBORG
UNIVERSITET

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Parallelizability of Problems

Parallel Program Development Process

Problem formulation

Algorithm design

Implementation

Optimization

Design of Parallel Programs

- Understand the problem (and algorithm)
  - Understand the problem to be solved in parallel
  - If starting with a sequential algorithm, its understanding is important
- Determine whether the problem is (actually) **parallelizable** → avoid time wastage
- Identify and focus on the **hotspots** → where is the most work done?
- Identification of **bottlenecks** → any disproportionately slow parts? e.g., input/output
- Identify **parallelism inhibitors** e.g., data dependence

Parallelizability of Problems

- Two extreme cases:
  - **embarrassingly/perfectly parallel problems** → little or no effort is needed to separate the problem into a number of parallel tasks
    - often the case where there is little or **no dependency** or **need for communication** between those parallel tasks, or for results between them
    - Example:
      - Monte Carlo simulations
      - Image processing
  - Inherently serial problems/**non-parallelizable** problems → cannot be split up into parallel portions
    - Calculate the Fibonacci series using a simple recursive approach:
- But what is in between these extreme cases?

Parallelizability of Problems

- Embarrassingly Parallel:
  - Perfectly parallel, no dependencies and tasks can run entirely independently.

- **Regular Parallel:**
  - Very high parallelism, but a few tasks may require some dependency or coordination.

- **Slightly Parallel:**
  - Some parallelism possible, but a significant portion of the task must be done sequentially.

- **Difficult to Parallelize:**
  - Requires complex synchronization and task management.

- Non-Parallelizable:
  - Cannot be parallelized due to inherent dependencies between tasks

Key Considerations for Parallelizability

- **Task Independence:**
  - How **independent** are the tasks from one another? → More independence means higher parallelizability.

- **Dependencies:**
  - How interdependent are the tasks? → More dependencies mean harder or less parallelizable.

- **Synchronization Costs:**
  - Does the problem require **frequent communication or coordination** between parallel tasks? → High synchronization costs reduce the effectiveness of parallelism.

- **Granularity:**
  - The size of the subproblems that can be parallelized → Finer granularity can increase parallelism, but it also increases overhead.

Design Paradigms

Binary Tree Paradigm

- A binary tree with n nodes is of height $\log_2(n)$
  - Suppose there are n data items, corresponding to the n leaf vertices of a **complete binary tree**
  - Process them in **parallel and get partial results** at the non-leaf nodes
  - Proceed bottom up, and reach the root in $\log_2(n)$ time

Example: **Sum of n numbers**

- It takes O(n) time for a single processor to compute the sum
- Assume $n = 2^d = 8$ and $p = \frac{n}{2} = 4$

- Suppose that the data to be added is

Item	A(1)	A(2)	A(3)	A(4)	A(5)	A(6)	A(7)	A(8)
Value	9	15	8	10	22	7	4	6

- Sketch the binary tree for a parallel program using the Binary Tree Paradigm

Binary Tree Paradigm

Algorithm **parallelSUM**

Input: Array $A[1, \dots, n]$, where $n = 2^d$

Output: The sum of the values in the array stored in $A[1]$

1: // computes sum of $A[1], A[2], \dots, A[n]$
2: $z = \frac{n}{2}$
3: **while** $z > 0$ **do**
4: **for** $id = 1 : z$ **do in parallel**
5: $A[id] = A[2 \times id - 1] + A[2 \times id]$
6: **End parallel**
7: $z = \left\lceil \frac{z}{2} \right\rceil$
8: **return** $A[1]$

Time complexity: $O(\log_2(n))$

Using $O(n)$ processors on an EREW PRAM architecture

Stage 1:

P_1 do $A(1) \leftarrow A(1) + A(2) = 24$
P_2 do $A(2) \leftarrow A(3) + A(4) = 18$
P_3 do $A(3) \leftarrow A(5) + A(6) = 29$
P_4 do $A(4) \leftarrow A(7) + A(8) = 10$

Stage 2:

P_1 do $A(1) \leftarrow A(1) + A(2) = 42$
P_2 do $A(2) \leftarrow A(3) + A(4) = 39$

Stage 3:

P_1 do $A(1) \leftarrow A(1) + A(2) = 81$

Growing By Doubling

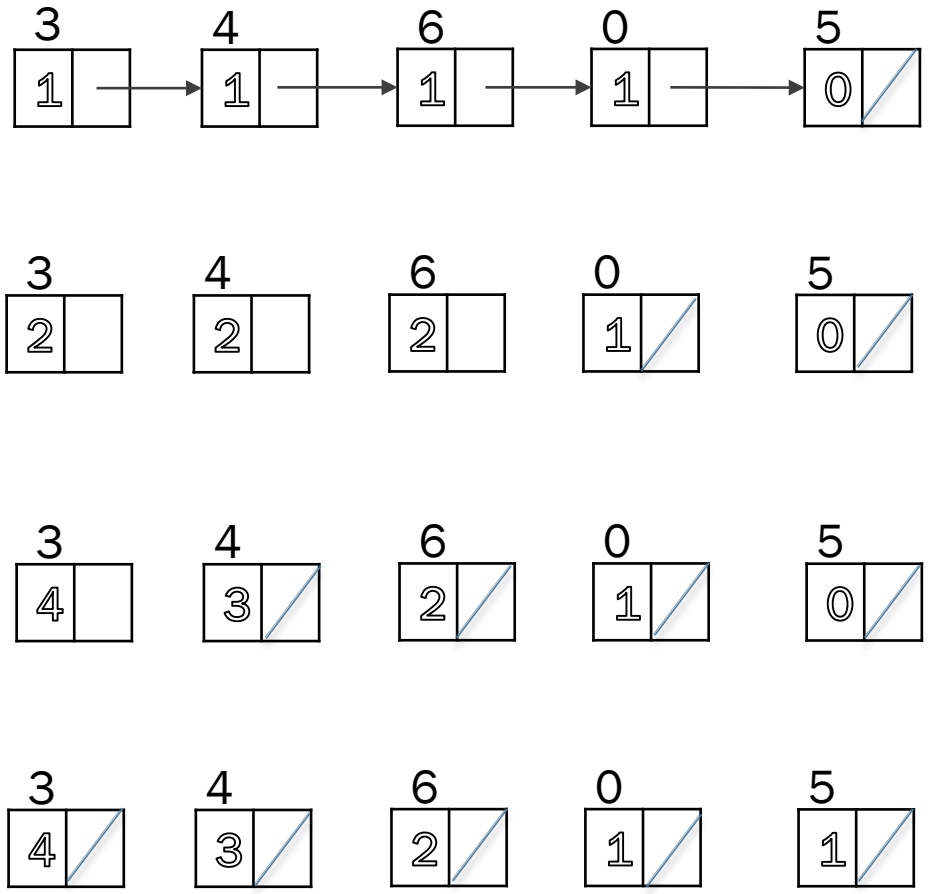
Concept:

- Doubling strategy is a common parallel programming technique where the **problem size or work units double** at each step.
  - At the initial step, each processor covers a single entity and waits for consolidation
  - At every step the number of entities covered is double the entities covered at the previous step: 2, 4, 8, etc.

- Example: List Ranking
  - Given a **linked list**, the goal is to assign a rank to each node, indicating its distance from the head.
  - Used in parallel computing for **graph algorithms**, **parallel prefix computation**, and **Euler tour** techniques.

List Ranking

- Sequential
  - Traverse each node one by one → $O(n)$ time
- Parallel (Growing by Doubling)
  - Nodes update their ranks in parallel → $O(\log_2(n))$ time
    - Pointer jumping steps
  - Update rule:
    - $Next(X) = Next\big(Next(X)\big)$
    - $Rank(X) = Rank(X) + Rank\big(Next(X)\big)$

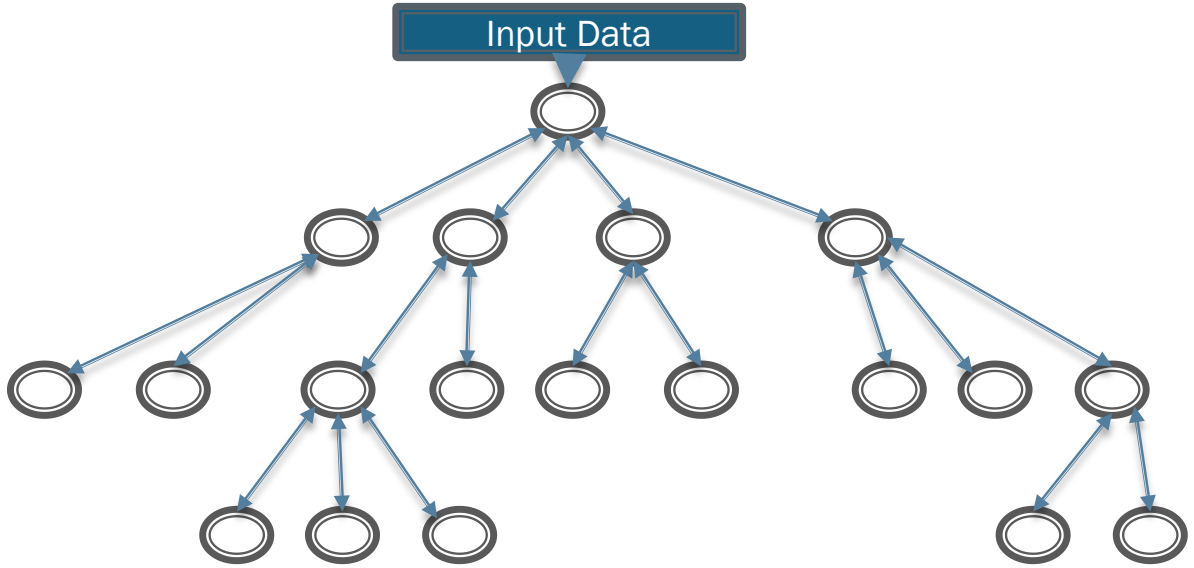


Divide and Conquer

- Divide and Conquer technique: If there is a direct solution to a given problem then it is applied directly. Otherwise the problem is repeated partitioned into subproblems.

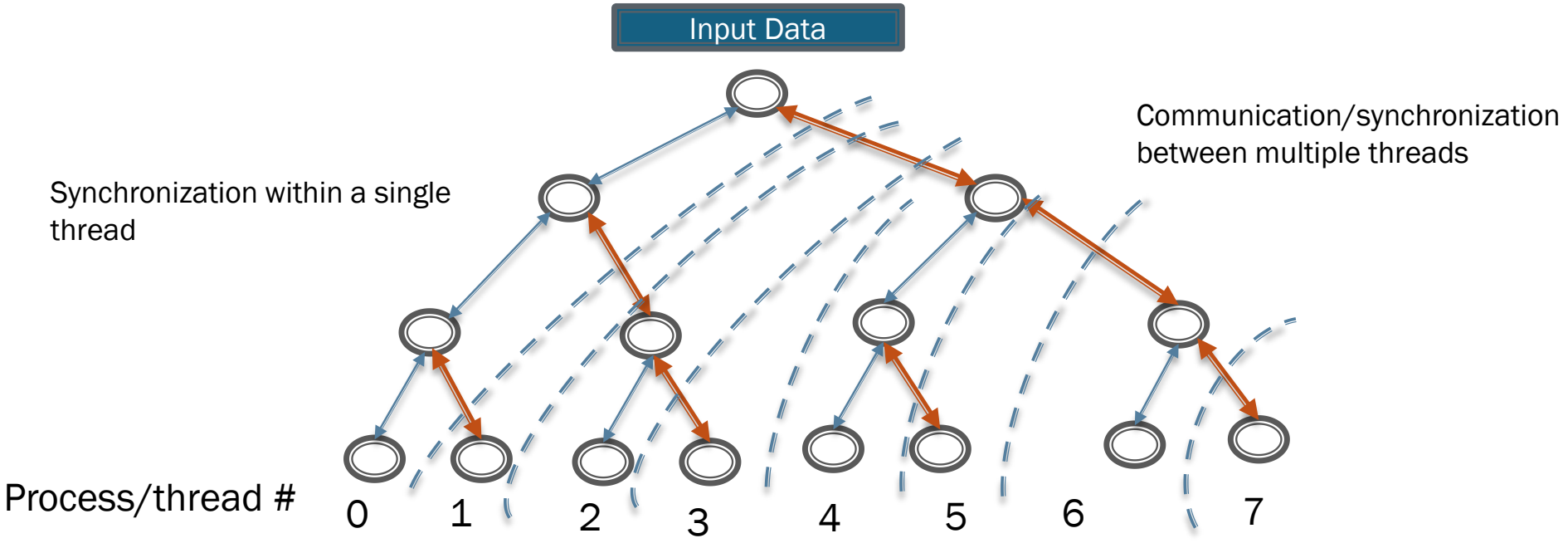
- A divide-and-conquer algorithm can be represented as a binary tree:
  - Characteristics:
    - Node degrees
    - Tree Depth

What makes good characteristics of a binary for divide-and-conquer?



An imbalanced tree illustration of divide-&-conquer processing

Parallelization of Divide-and-conquer Tree



- A divide-and-conquer application can be very dynamic in terms of:
  - Dependency of subtree generation on previously computed results
  - Varying time for processing nodes, especially leaves of the tree.

- A dynamic approach is neccessary for load balancing.

Parallel Algorithm Design Process

Parallel Algorithm Design Principle

- A given programming problem may have multiple parallel algorithmic solutions
  - Best solution may not be the one suggested by existing sequential algorithms

- General design process:
  - **focus on machine-independent issues** (concurrency, parallelism, etc)
  - Machine dependent considerations are left for later part of the design process

- Four stage design process:
  - **Decomposition** → Communication → Agglomeration/Clustering → Mapping

Decomposition (or Partitioning)

- Decompose the problem into tasks that can be executed concurrently
  - show opportunities for parallel execution
  - obtain a fine-grained decomposition of the problems → large number of small tasks
- A good partition divides into small pieces both the computation associated with a problem and the data on which this computation operates
  - A given problem may be decomposed into tasks in many ways.
- **Two possible approaches:**

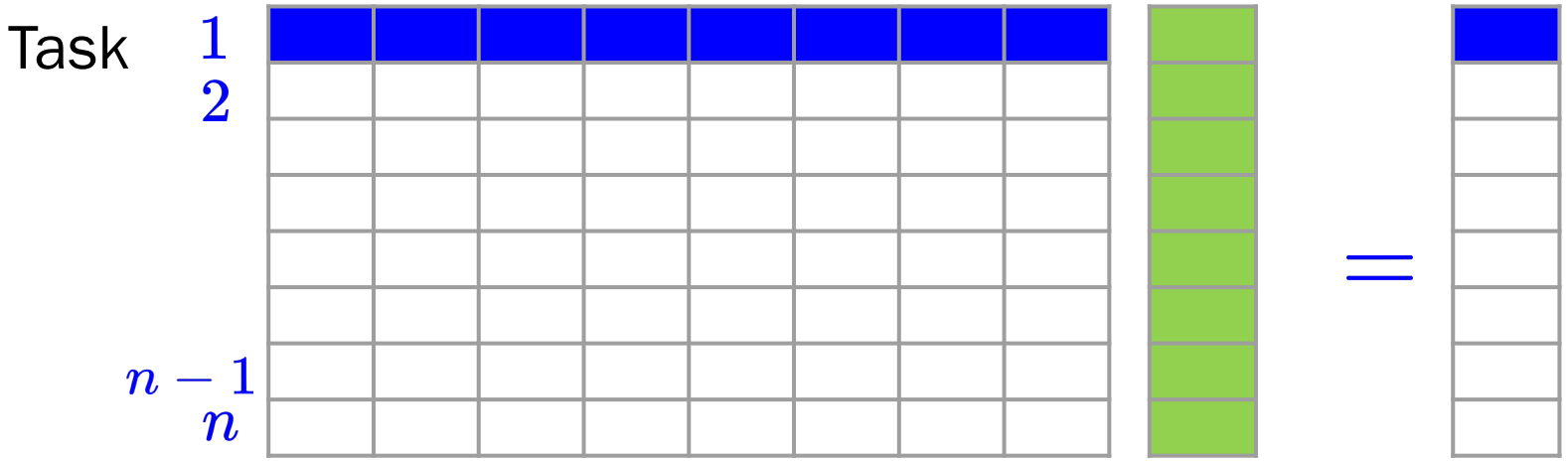
  1. **Domain decomposition:** data partitioning first and then associated computations
  2. **Function decomposition:** start with portioning of the computations and then data

A decomposition can be illustrated in the form of a **directed graph**

- nodes corresponding to tasks
- edges indicating that the result of one task is required for processing the next.
- Such a graph is called a **task dependency graph or computation graph or ..... .**

Example: Dense Matrix-vector Multiplication

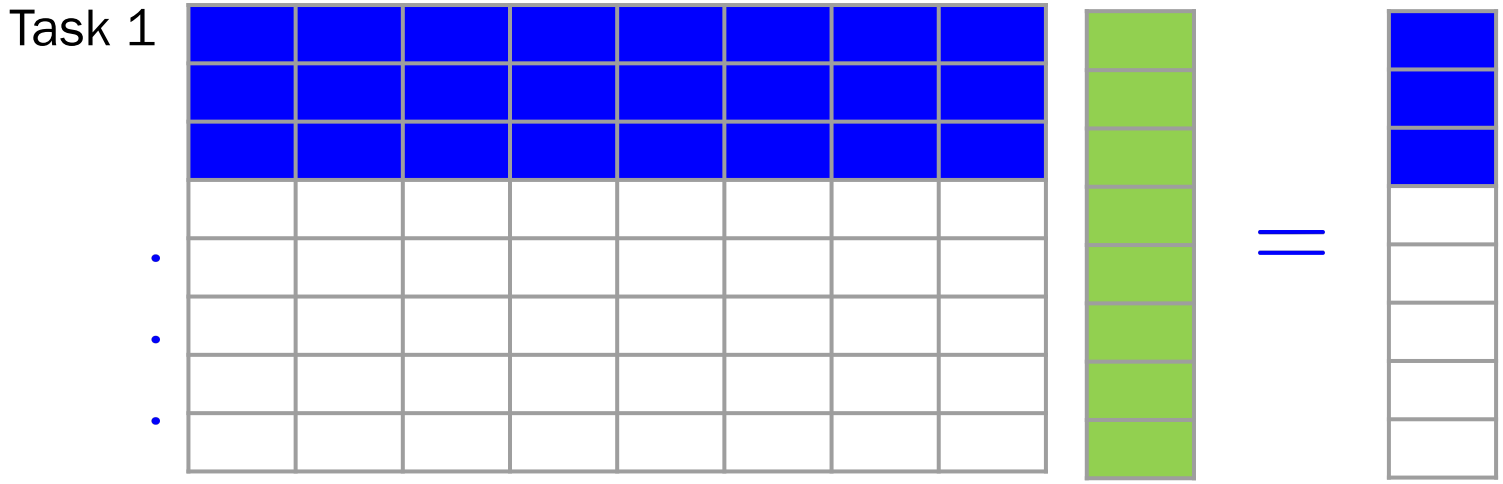
- Consider the matrix vector multiplication: $y = A \times b$



- Computation of each element of output vector y is independent of other elements.
  - a dense matrix-vector product can be decomposed into n tasks.
- Observations:
  - While tasks share data (i.e., b), they do not have any control dependencies - i.e., no task needs to wait for the (partial) completion of any other.
  - All tasks are of the same size in terms of number of operations.
  - Is this the maximum number of tasks we could decompose this problem into?

Granularity of Task Decompositions

- The number of tasks into which a problem is decomposed determines its **granularity**.
- Decomposition into:
  - a large number of tasks results in **fine-grained decomposition**
  - a small number of tasks results in a **coarse-grained decomposition**.



A coarse grained counterpart to the dense matrix-vector product example.

- Each task in this example corresponds to the computation of three elements of the result vector.

Data (Domain) Decomposition: Example

- Consider the problem of multiplying two $n \; x \; n$ matrices A and B to yield matrix C.
- The output matrix C can be partitioned into four tasks as follows (or even 8):

$$\begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix} \cdot \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix} \rightarrow \begin{pmatrix} C_{1,1} & C_{1,2} \\ C_{2,1} & C_{2,2} \end{pmatrix}$$

$$C_{1,1} = A_{1,1}B_{1,1} + A_{1,2}B_{2,1}$$

$$C_{1,2} = A_{1,1}B_{1,2} + A_{1,2}B_{2,2}$$

$$C_{2,1} = A_{2,1}B_{1,1} + A_{2,2}B_{2,1}$$

$$C_{2,2} = A_{2,1}B_{1,2} + A_{2,2}B_{2,2}$$

Decomposition I	Decomposition II
Task 1: $C_{1,1} = A_{1,1} B_{1,1}$	Task 1: $C_{1,1} = A_{1,1} B_{1,1}$
Task 2: $C_{1,1} = C_{1,1} + A_{1,2} B_{2,1}$	Task 2: $C_{1,1} = C_{1,1} + A_{1,2} B_{2,1}$
Task 3: $C_{1,2} = A_{1,1} B_{1,2}$	Task 3: $C_{1,2} = A_{1,2} B_{2,2}$
Task 4: $C_{1,2} = C_{1,2} + A_{1,2} B_{2,2}$	Task 4: $C_{1,2} = C_{1,2} + A_{1,1} B_{1,2}$
Task 5: $C_{2,1} = A_{2,1} B_{1,1}$	Task 5: $C_{2,1} = A_{2,2} B_{2,1}$
Task 6: $C_{2,1} = C_{2,1} + A_{2,2} B_{2,1}$	Task 6: $C_{2,1} = C_{2,1} + A_{2,1} B_{1,1}$
Task 7: $C_{2,2} = A_{2,1} B_{1,2}$	Task 7: $C_{2,2} = A_{2,1} B_{1,2}$
Task 8: $C_{2,2} = C_{2,2} + A_{2,2} B_{2,2}$	Task 8: $C_{2,2} = C_{2,2} + A_{2,2} B_{2,2}$

- Unique decomposition of tasks is not guaranteed.

Communication

- Tasks from problem decomposition are intended to be executed concurrently
  - concurrency does not generally imply independence
  - operations performed on one task typically require data associated with another task(s)

- The communication phase of the design involve defining how data is transferred between multiple concurrent task executions

- Communication in parallel processing can be seen as a two stage process:
  - specify communication channels between tasks (considering topology)
  - define the type of messages to be sent on the channels

Goal: optimize performance by distributing communication operations over many tasks and by organizing communication operations in a way that permits concurrent execution.

- minimize number of channels and communication requirements.

Distribution of Communication and Computation

- We return to the compute sum algorithm for finding the summation of N numbers, i.e.,

$$S = \sum_{n=1}^{N} X_n$$

- We can distribute the computation into N tasks such that the nth task compute

$$S_n = X_n + S_{n-1}$$

- The compute sum algorithm
  - distributes the N-1 communication and additions
  - permits concurrent execution **only** if multiple operations are to be performed

How do we uncover concurrency?

- To solve a complex problem, we use divide and conquer
  - recursively partition the problem into two or more subproblems
  - solve subproblems and combine solutions

- For the summation problem
  - Using the identity: $N = 2^n$, we can decompose the problem into two:

$$\sum_{i=0}^{2^n-1} X_i = \sum_{i=0}^{2^{n-1}-1} X_i + \sum_{i=2^{n-1}-1}^{2^n-1} X_i$$

Binary Tree Representation

How does the divide and conquer algorithm compare with the linear array summation in the previous slide?

How do we access communication design for a parallel algorithm?

- **Do all tasks perform about the same number of communication operations?**
  - Unbalanced communication requirements suggest a non-scalable construct.
  - Revisit your design to see whether communication operations can be distributed more equitably.

- **Does each task communicate only with a small number of neighbors?**
  - If each task must communicate with many other tasks, evaluate the possibility of formulating this global communication in terms of a local communication structure.

- **Are communication operations able to proceed concurrently?** If not, your algorithm is likely to be inefficient and non-scalable.

- **Is the computation associated with different tasks able to proceed concurrently**? If not, your algorithm is likely to be inefficient and non-scalable. Consider whether you can reorder communication and computation operations.

Agglomeration/Clustering

- So far:
  - Decomposition: Partition computation into a set of tasks
  - Communication: design channels to provide data to the tasks

- But the design is still rather abstract → no specialized mechanism for execution on any parallel computer.

- Shifting a bit from abstraction to more concreteness → obtaining parallel algorithms to run on specific class or classes of parallel computers
  - revisit decomposition and communication design choices
  - consider combining tasks to provide a reduced number of larger tasks
  - determine whether algorithm may benefit from replication of data and/or computation

Processes and Mapping

- In general, the number of tasks in a decomposition exceeds the number of processing elements.
  - a parallel algorithm must also provide a mapping of tasks to processes.

- We refer to the mapping as being from **tasks to processes**, as opposed to processors.
  - typical programming APIs do not allow easy binding of tasks to physical processors.
  - we aggregate tasks into processes and rely on the system to map these processes to physical processors.
  - processes here simply refer to a **collection of tasks and associated data.**

- Appropriate mapping of tasks to processes is critical to **parallel performance**
  - Mappings are determined by both the **task dependency** and **task interaction** graphs.
  - Task dependency graphs can be used to ensure that work is equally spread across all processes at any point (**minimum idling and optimal load balance**).

Processes and Mapping

- Task interaction graphs can be used to make sure that processes need minimum interaction with other processes (minimum communication).

- An appropriate mapping must minimize parallel execution time by:
  - Mapping independent tasks to different processes.
  - Assigning tasks on **critical path** to processes as soon as they become available.

- Minimizing interaction between processes by mapping tasks with dense interactions to the same process.

Note: These criteria often conflict each other. For example, a decomposition into one task (or no decomposition at all) minimizes interaction but does not result in a speedup at all!

Can you think of other such conflicting cases?

Characteristic of Tasks and Interactions

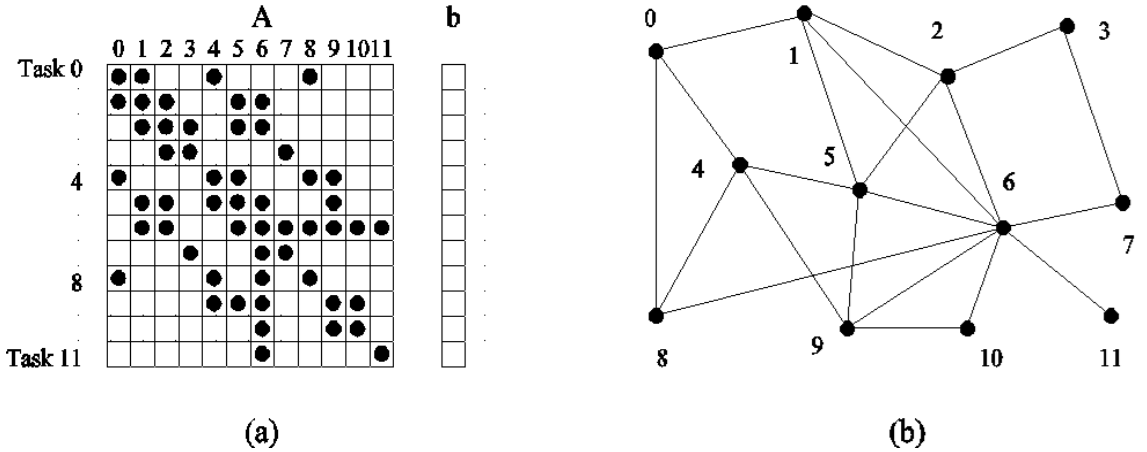
Task Interaction Graphs

- Subtasks generally exchange data with others in a decomposition.
  - The graph of tasks (nodes) and their interactions/data exchange (edges) is referred to as a task interaction graph.
  - task interaction graphs represent data dependencies, whereas task dependency graphs represent control dependencies.

Example: Consider the problem of multiplying a sparse matrix A with a vector b.

  - the computation of each element of the result vector can be viewed as an independent task.
  - only non-zero elements of matrix A participate in the computation.

- If, for memory optimality, we also partition b across tasks, then the task interaction graph of the computation is identical to the graph of the matrix A (the graph for which A represents the adjacency structure).



Characteristics of Task Interactions

- Tasks may communicate with each other in various ways:
  - Static interactions: The tasks and their interactions are known a-priori.
    - These are relatively simpler to code into programs.
  - Dynamic interactions: The timing or interacting tasks cannot be determined a-priori.
    - These interactions are harder to code, especially, as we shall see, using message passing APIs.
  - Regular interactions: There is a definite pattern (in the graph sense) to the interactions.
    - These patterns can be exploited for efficient implementation.
  - Irregular interactions: Interactions lack well-defined topologies.
- Interactions may be read-only or read-write.
  - In read-only interactions, tasks just read data items associated with other tasks.
  - In read-write interactions tasks read, as well as modify data items associated with other tasks.
- Interactions may be one-way or two-way.
  - A one-way interaction can be initiated and accomplished by one of the two interacting tasks.
  - A two-way interaction requires participation from both tasks involved in an interaction.
  - One way interactions are somewhat harder to code in message passing APIs.

Degree of Concurrency

- The number of tasks that can be executed in parallel is the degree of concurrency of a decomposition.

- Since the number of tasks that can be executed in parallel may change over program execution,
  - the maximum degree of concurrency is the maximum number of such tasks at any point during execution.
  - What is the maximum degree of concurrency of the summation example?

- The **average degree of concurrency** is the average number of tasks that can be processed in parallel over the execution of the program.

- The degree of concurrency increases as the decomposition becomes finer in granularity and vice versa.

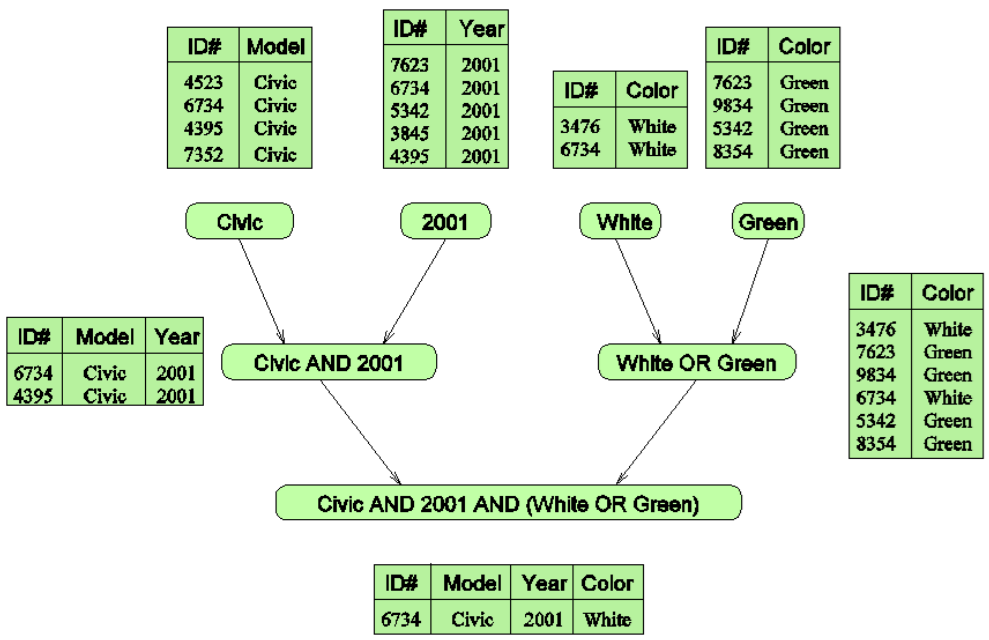
Example: Database Query Processing

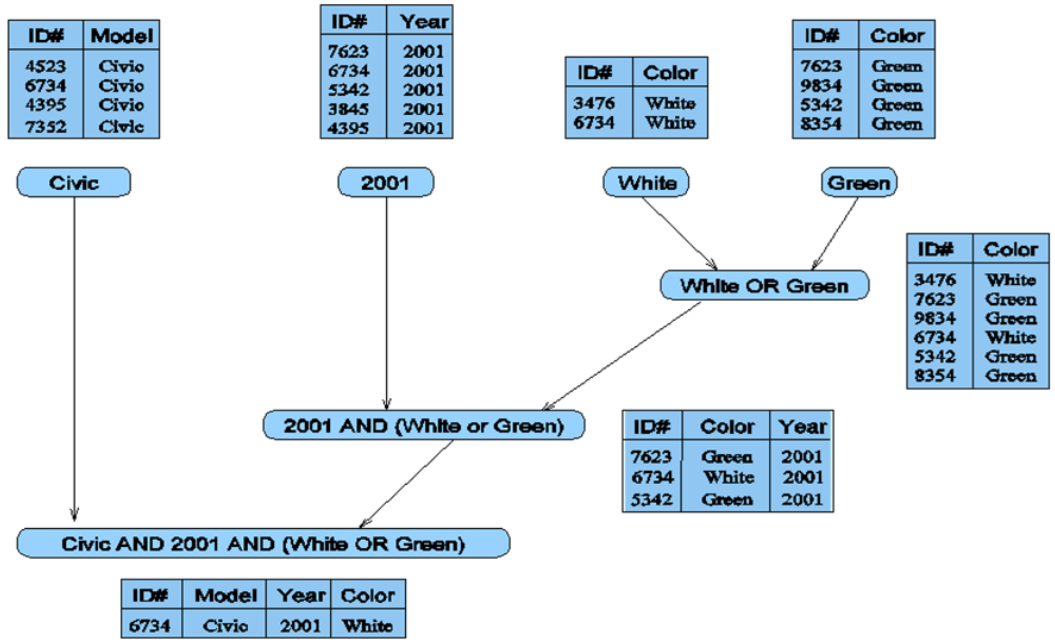
Consider the execution of the query: *MODEL = ``CIVIC'' AND YEAR = 2001 AND (COLOR = ``GREEN'' OR COLOR = ``WHITE)* on the following database:

ID#	Model	Year	Color	Dealer	Price
4523	Civic	2002	Blue	MN	$18,000
3476	Corolla	1999	White	IL	$15,000
7623	Camry	2001	Green	NY	$21,000
9834	Prius	2001	Green	CA	$18,000
6734	Civic	2001	White	OR	$17,000
5342	Altima	2001	Green	FL	$19,000
3845	Maxima	2001	Blue	NY	$22,000
8354	Accord	2000	Green	VT	$18,000
4395	Civic	2001	Red	CA	$17,000
7352	Civic	2002	Red	WA	$18,000

Example: Database Query Processing

- The execution of the query can be divided into subtasks in various ways. Each task can be thought of as generating an intermediate table of entries that satisfy a particular clause.





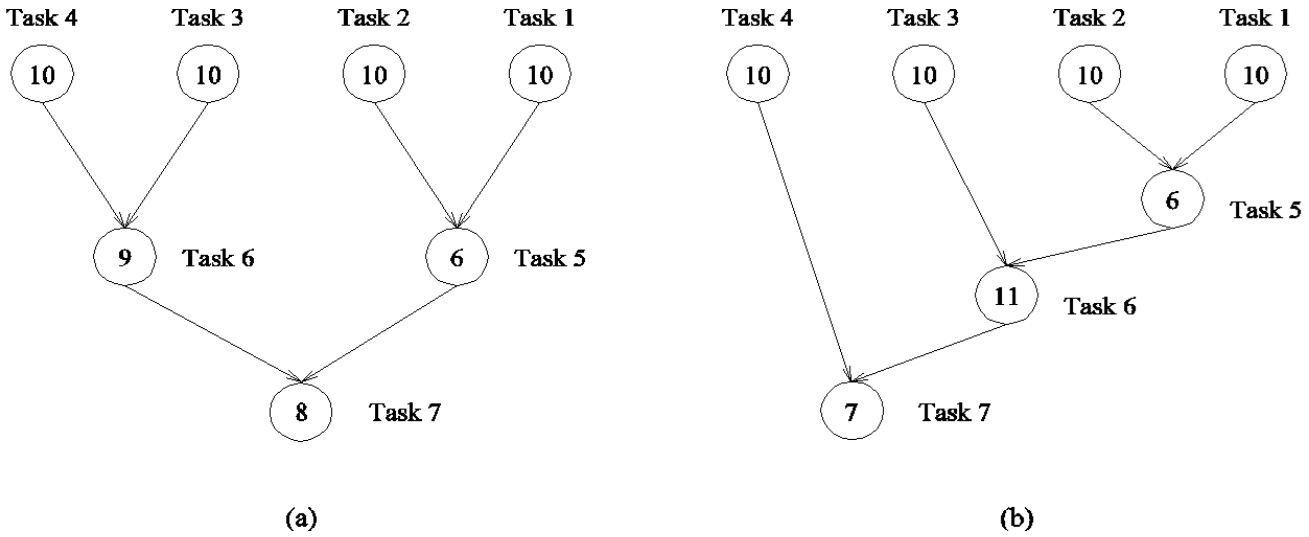
- Two possible decomposition of the given query into a number of tasks. Edges in this graphs denote that the output of one task is needed to accomplish the next.

Critical Path Length

A directed path in the task dependency graph represents a sequence of tasks that must be processed one after the other.

- The longest such path determines the shortest time in which the program can be executed in parallel.
- The length of the longest path in a task dependency graph is called the **critical path length**.

Consider the task dependency graphs of the two database query decompositions:



Property	a	b
Max deg. of conc.		
Av. deg of conc.		
Critical path length		

How do we map these task to processes (assuming a maximum of 4)?

Limits on Parallel Performance

- It would appear that the parallel time can be made arbitrarily small by making the decomposition finer in granularity.
- But there is an inherent bound on how fine the granularity of a computation can be.

- For example, in the case of multiplying a dense matrix with a vector, there can be no more than (n^2) concurrent tasks.

- Concurrent tasks may also have to exchange data with other tasks → communication overhead.

- The tradeoff between the **granularity of a decomposition** and associated **overheads** often determines performance bounds.

Summary and Exercise

Summary

- Not all problems are parallelizable
  - Pay careful attention to understanding the problem and its parallelizability before venturing into algorithm design
- Several paradigms exist for analysis and design of parallel algorithms
  - Binary tree, growing by doubling, parallel divide-and-conquer, partitioning
    - Use of each paradigm depends on the type of problem and its parallelizability
- Design of a parallel algorithm for any given problem can be divided into 4 stages:
  - Decomposition/partitioning → communication design → agglomeration → mapping
- Task dependency and task interaction graphs are useful for analysing communication and the mapping of tasks to processes

- Even for parallelizable problems, there is a limit on the granularity of decomposition
  - Consider how fine or grained the granularity should be for best performance.

Summary

- Two sometimes conflicting useful strategies for mapping
  - place tasks that can execute concurrently on different processors to enhance concurrency.
  - place tasks that communicate frequently on the same processor, to increase locality.

- Typical algorithms rely on domain decomposition
  - Efficient in cases featuring equal size tasks and structured communication
  - May be difficult on applications with varied work per task and/or unstructured communication
  - Use load balancing to identify efficient agglomeration and mapping techniques

That’s it for today!